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Aryabhatta Knowledge University, Patna

B.Tech. 5th Semester Examination, 2019

Time: 03 Hours

Code: 051516 ***Subject: Compiler Design*** ***Full Marks: 70***

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 Choose the correct answer of the following (any seven): **[14]**

(a) When is the type checking usually done?
- (i) During syntax directed translation
- (ii) During lexical analysis
- (iii) During code optimization
- (iv) During syntax analysis

(b) Which one of the following statements is true?
- (i) Canonical LR parser is more powerful than LALR parser.
- (ii) SLR parser is more powerful than LALR.
- (iii) LALR parser is more powerful than canonical LR parser.
- (iv) SLR parser, canonical LR parser and LALR parser all have the same power.

(c) In a compiler, _____ checks every character of the source text.
- (i) the lexical analyzer
- (ii) the syntax analyzer
- (iii) the code generator
- (iv) the code optimizer

(d) For a grammar G, shift reduce (S-R) conflicts are present in LALR(1) parser, if and only if
- (i) the LR(1) parser for G has S-R conflicts
- (ii) the LR(0) parser for G has S-R conflicts
- (iii) the SLR(1) parser for G has S-R conflicts
- (iv) the SLR(0) parser for G has S-R conflicts

(e) In an absolute loading scheme, which loader function is accomplished by programmer?
- (i) Allocation
- (ii) Linking
- (iii) Reallocation
- (iv) Both (i) and (ii)

(f) _____ is a top-down parser.
- (i) Operator precedence parser
- (ii) An LALR (k) parser
- (iii) An LR (k) parser
- (iv) Recursive descent parser

(g) The languages that need heap allocation in the runtime environment are those that
- (i) use global variables
- (ii) use dynamic scoping
- (iii) support recursion
- (iv) allow dynamic data structure

(h) In compilers, generation of intermediate code based on an abstract machine model is useful because
- (i) syntax-directed translations can be written for intermediate code generation
- (ii) to generate code for real machines directly from high-level language program is not possible
- (iii) portability of the front end of the compiler is enhanced
- (iv) implementation of lexical and syntax analyses is easier

(i) To convert an arbitrary CFG to an LL(1) grammar
- (i) factor the grammar alone
- (ii) remove left recursion alone
- (iii) remove left recursion and factor the grammar
- (iv) None of the above

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(j) The method which merges the bodies of two loops is

(i) loop rolling
(ii) loop jamming
(iii) constant folding
(iv) None of the above

Q.2 Solve both questions : **[14]**

(a) What is an activation record? Explain how they are used to access various local and global variables.

(b) What is bottom-up parsing? Discuss shift reduce parsing technique in brief. What is a handle?

Q.3 Solve both questions : **[14]**

(a) What is left recursion? Eliminate the left recursion from the following grammar:
$E \rightarrow E+T \mid T$
$T \rightarrow T*\mathrm{F} \mid \mathrm{F}$
$\mathrm{F} \rightarrow (E) \mid id$

(b) What is the use of a symbol table? How are the identifiers stored in the symbol table?

Q.4 Solve both questions : **[14]**

(a) What is the pass of a compiler? Explain how the single- and multi-pass compilers work.

(b) Explain architecture and algorithm for the non-recursive predictive parser.

Q.5 Solve both questions : **[14]**

(a) Check whether the following grammar is CLR or not:
$S \rightarrow Aa \mid bBa \mid Ba \mid bAc$
$\mathrm{A} \rightarrow c$
$B \rightarrow d$

(b) What is the syntax directed translation and why are they important?

Q.6 Solve both questions : **[14]**

(a) Explain different phases of compiler.

(b) Explain how type checking and error reporting are performed in compiler. Draw syntax tree and DAG for the statement:
$a=(a*b+c)\text{wedge}(b+c)*b+c$

Q.7 Solve both questions : **[14]**

(a) Explain various targets for code optimization with examples.

(b) How are CPU registers allocated while creating machine code?

Q.8 Solve both questions : **[14]**

(a) Check whether the following grammar is LL(1) grammar or not:
$S \rightarrow iEtSA \mid a$
$\mathrm{A} \rightarrow eS \mid \varepsilon$
$E \rightarrow b$

(b) Design the FIRST SET and FOLLOW SET for the grammar.

Q.9 Solve both questions : **[14]**

(a) Differentiate between S-attribute SDT and L-attribute SDT with suitable examples.

(b) Explain any two of the following:

(i) Lexical phase error
(ii) Code generation using dynamic programming
(iii) Syntax tree

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